238531: metamonzogranite, Scotties Bore

(Big Bell Suite, Austin Downs Supersuite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Cue (SG 50-15), Dalgaranga (2342)
MGA Zone 50, 513733E 6933814N

Sampled on 13 October 2017

This sample was collected from outcrop on the east side of a track, about 10.1 km east-southeast of Buongnoo Hill, 3.4 km south of Scotties Bore, and 0.7 km north of the Mount Farmer Road.

Geological context

The unit sampled is a gneissic metamonzogranite of the Big Bell Suite of the Austin Downs Supersuite (Van Kranendonk et al., 2013; Zibra, 2019). The 2732–2686 Ma Big Bell Suite consists mainly of porphyritic and equigranular monzogranite and subordinate granodiorite and tonalite, together with their deformed and metamorphosed equivalents. The Big Bell Suite discordantly intruded mafic to felsic igneous and metasedimentary rocks of the Norie, Polelle and Glen Groups, as well as metagranitic and gabbroic rocks of the Annean Supersuite, and is intruded by granitic rocks of the Tuckanarra and Jungar Suites, and the Bald Rock Supersuite (Van Kranendonk et al., 2013; Zibra, 2019). Most Big Bell Suite granitic rocks exhibit a well-developed, north- to northeast-trending magmatic foliation, subparallel to the regional granite–greenstone fabric (Zibra, 2019). At this locality, the metamonzogranite displays a north-trending gneissic fabric, and is crosscut by metre-scale monzogranite dykes. A monzogranite collected about 41 km to the east-southeast yielded an igneous crystallization age of 2702 ± 4 Ma (GSWA 155503, Wingate et al., 2014b), and is crosscut by a granitic dyke dated at 2686 ± 4 Ma (GSWA 155502, Wingate et al., 2014a). A porphyritic monzogranite collected about 51 km to the southeast yielded an igneous crystallization age of 2701 ± 2 Ma (GSWA 203717, Wingate et al., 2019b), and is intruded by the 2692 ± 2 Ma Boogardie Orbicular Granite (GSWA 203716, Wingate et al., 2019a; GSWA 203721, Wingate et al., 2019c; Wingate, 2020).

Petrographic description

The sample is a medium-grained, equigranular metamonzogranite (Fig. 1), consisting of about 40% plagioclase, 25–30% K-feldspar, 25–30% quartz, 5% biotite, 2% myrmekite, <1% magnetite, and accessory muscovite, apatite and zircon. Plagioclase is anhedral to subhedral, equant to slightly tabular, up to 5 mm in size, and variably sericitized. K-feldspar is finely microperthitic, and forms subhedral, tabular crystals up to 6 mm long, and anhedral grains up to 0.5 mm long in polygonal aggregates with fine-grained quartz and plagioclase. Some larger K-feldspar grains contain inclusions of other minerals, many exhibit microcline twinning, and some are bordered by minor myrmekite. Quartz is irregular, anhedral, up to 4 mm in diameter, finely fractured, and exhibits weak undulose extinction. Biotite forms mainly subhedral laths up to 1.5 mm long, many in decussate aggregates, and is partially altered to chlorite. Muscovite forms subhedral, randomly oriented laths up to 0.3 mm long, and anhedral magnetite up to 0.7 mm in size is partly replaced by hematite along fractures. Apatite is subhedral and up to 0.3 mm long. Undulose extinction and fine fracturing in quartz suggests the sample may have been deformed and partially recovered, probably at moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are pale brown to dark brown or opaque, and mainly subhedral. The crystals are up to 350 µm long, and elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, most crystals are dominated by high-U, metamict domains, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 20–21 and 26 September 2019, using SHRIMP-B. Analyses 1.1 to 8.1 (spot numbers 1–8) were obtained during the first session, together with six analyses of the M257 standard, of which five analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.59% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.35% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.71%. Analyses 9.1 to 11.1 (spot numbers 9–11) were obtained during the second session, together with four analyses of the M257 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.67% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

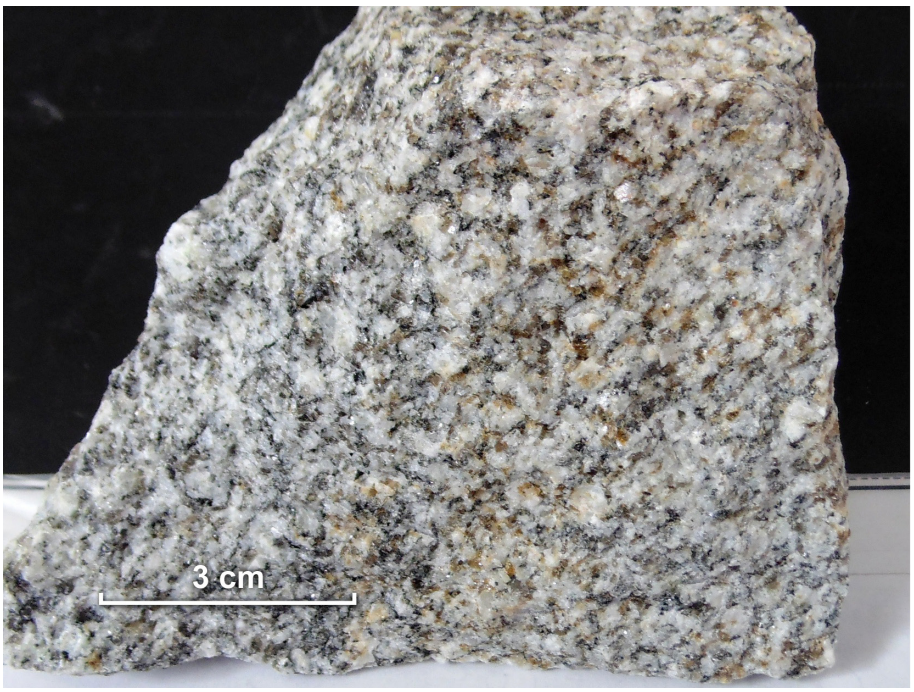

Figure 1. Hand specimen image of sample 238531: metamonzogranite, Scotties Bore

Figure 2. Cathodoluminescence image of representative zircons from sample 238531: metamonzogranite, Scotties Bore. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 238531: metamonzogranite, Scotties Bore

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	10	10.1	56	22	0.39	0.589	1.938	0.038	0.18978	0.00381	1.950	0.038	0.18453	0.00418	2669	44	2694	37	0.9
I	9	9.1	602	77	0.13	0.131	1.938	0.020	0.18582	0.00062	1.941	0.020	0.18465	0.00066	2679	24	2695	6	0.6
I	6	6.1	440	95	0.22	0.123	1.957	0.016	0.18653	0.00050	1.959	0.016	0.18544	0.00055	2659	19	2702	5	1.6
I	3	3.1	182	101	0.55	0.100	1.882	0.018	0.18720	0.00076	1.884	0.018	0.18632	0.00080	2745	22	2710	7	-1.3
I	2	2.1	120	73	0.61	0.089	1.888	0.020	0.18731	0.00092	1.889	0.020	0.18652	0.00096	2738	25	2712	9	-1.0
I	8	8.1	349	185	0.53	0.047	1.967	0.021	0.18710	0.00057	1.968	0.021	0.18669	0.00058	2649	24	2713	5	2.4
I	7	7.1	104	96	0.92	0.070	1.945	0.031	0.18737	0.00103	1.947	0.031	0.18676	0.00107	2672	35	2714	9	1.5
I	5	5.1	590	514	0.87	0.082	1.915	0.015	0.18785	0.00043	1.917	0.015	0.18712	0.00046	2707	18	2717	4	0.4
X	11	11.1	206	180	0.87	0.310	1.918	0.036	0.19201	0.00109	1.923	0.036	0.18924	0.00125	2699	43	2736	11	1.3
X	1	1.1	148	101	0.68	0.089	1.758	0.018	0.20411	0.00084	1.760	0.018	0.20333	0.00087	2901	24	2853	7	-1.7
X	4	4.1	160	108	0.67	0.031	1.761	0.018	0.20546	0.00130	1.761	0.018	0.20519	0.00131	2899	24	2868	10	-1.1

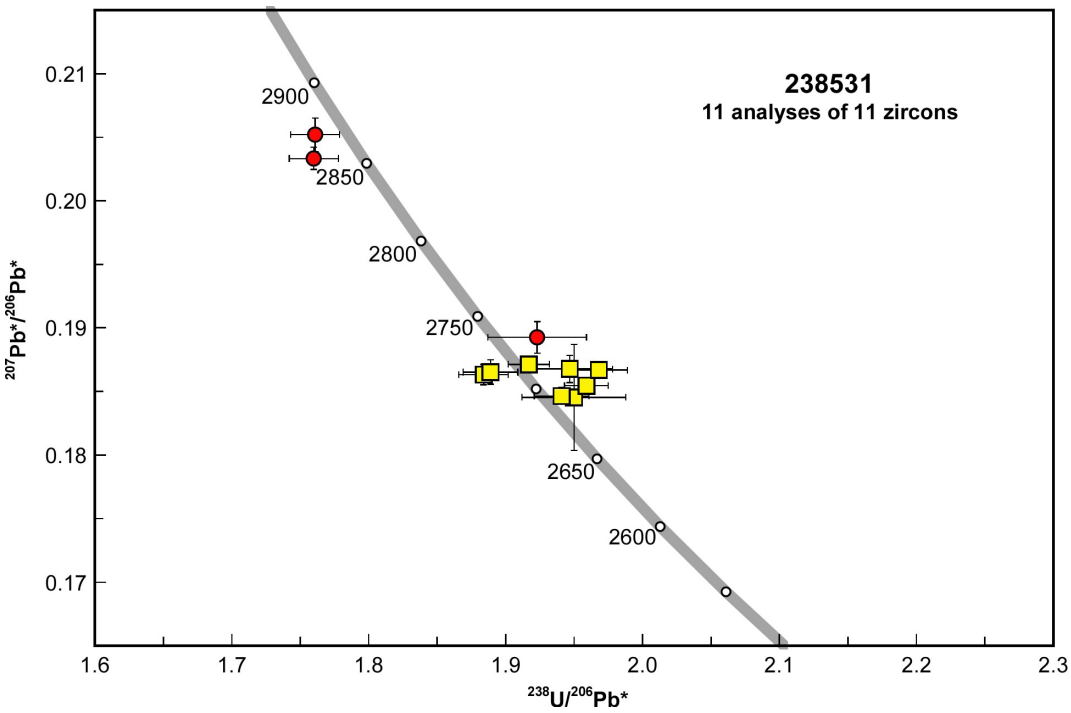

Figure 3. U–Pb analytical data for sample 238531: metamonzogranite, Scotties Bore. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons)

Results

Eleven analyses were obtained from 11 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their locations within the crystals.

Group I comprises eight analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2709 ± 7 Ma (MSWD = 1.8).

Group X comprises three analyses of zircon cores (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2868–2736 Ma.

The date of 2709 ± 7 Ma for the eight analyses in Group I is interpreted as the igneous crystallization age of the monzogranite. The dates of 2868–2736 Ma for the three analyses in Group X are interpreted as the ages of inherited zircons.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Zibra, I 2021, 238531: metamonzogranite, Scotties Bore; Geochronology Record 1804: Geological Survey of Western Australia, 4p.

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Data released: 21 October 2021